

Technical Note: Indirect Inference for Structural Econometric Models

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Abstract

Indirect inference is a simulation-based estimator for structural models that are easy to simulate but difficult to estimate by direct likelihood methods. The method proceeds through an auxiliary model: estimate a tractable statistical representation on the observed data, simulate the structural model at candidate parameter values, estimate the same auxiliary representation on the simulated data, and choose structural parameters that make the two sets of auxiliary estimates close. This note gives a pedagogical but technically explicit treatment of the method: the formal setup, the binding function, the Wald, likelihood-ratio, and score versions of the estimator, identification, asymptotic logic, weighting, simulation error, links to the simulated method of moments, and implementation diagnostics.

1 Purpose and basic intuition

Many structural models in economics imply a well-defined data-generating process but do not deliver a tractable likelihood. Examples include nonlinear dynamic models, heterogeneous-agent models, dynamic discrete-choice models, models with latent state variables, and equilibrium models with strategic interaction. In such settings, one can often simulate a panel or time series for any candidate structural parameter vector, even when the likelihood or score cannot be evaluated in closed form. Indirect inference uses this simulation ability to estimate the structural parameters (Smith, 1993; Gouriéroux et al., 1993).

The method has three ingredients:

1. a structural model indexed by parameters $\beta \in \mathcal{B} \subset \mathbb{R}^k$;
2. an auxiliary model indexed by parameters $\theta \in \Theta \subset \mathbb{R}^p$;
3. a distance criterion comparing auxiliary estimates from observed and simulated data.

The auxiliary model is not required to be correctly specified. Its role is to select informative features of the data. The structural estimator chooses β so that those features, when computed from simulated data, reproduce the same features in the observed data.

A compact description is:

$$\text{observed data} \xrightarrow{\text{auxiliary estimator}} \hat{\theta}, \quad \text{simulated data under } \beta \xrightarrow{\text{same auxiliary estimator}} \tilde{\theta}(\beta),$$

then choose β to minimize a weighted distance between $\hat{\theta}$ and $\tilde{\theta}(\beta)$.

2 Structural model

Let the observed sample be $\{(y_t, x_t) : t = 1, \dots, T\}$, where y_t is endogenous and x_t is observed and exogenous or predetermined. A canonical dynamic structural model is

$$y_t = G(y_{t-1}, x_t, u_t; \beta), \quad t = 1, \dots, T, \tag{1}$$

where u_t is an unobserved shock and $\beta \in \mathcal{B} \subset \mathbb{R}^k$ is the parameter vector of interest. The initial condition y_0 is taken as fixed or is drawn from a specified initialization rule. The shocks are drawn from a known distribution, say

$$u_t \sim F_0,$$

or from a parametric shock law whose parameters are included in β .

The key computational requirement is not that the likelihood of (1) be tractable. It is enough that, for each candidate β , the econometrician can generate artificial samples

$$\{\tilde{y}_t^m(\beta), x_t : t = 1, \dots, T\}, \quad m = 1, \dots, M,$$

using simulated shocks $\{\tilde{u}_t^m\}_{t=1}^T$. The superscript m denotes an independent simulation path. The observed exogenous sequence $\{x_t\}$ is often held fixed across simulations so that the simulated sample reproduces the observed design.

For numerical stability, the simulation shocks are usually drawn once and then held fixed across parameter evaluations. This “common random numbers” device turns part of the simulation noise into a fixed function of β and often makes the objective smoother.

3 Auxiliary model

The auxiliary model is a tractable econometric representation of selected data features. In likelihood form, suppose it is described by a conditional density

$$f(y_t \mid y_{t-1}, x_t; \theta), \quad \theta \in \Theta \subset \mathbb{R}^p. \quad (2)$$

The auxiliary parameter dimension p should be at least as large as the structural parameter dimension k for point identification in the usual overidentified or exactly identified cases:

$$p \geq k.$$

The observed-data auxiliary estimator is

$$\hat{\theta}_T = \arg \max_{\theta \in \Theta} \frac{1}{T} \sum_{t=1}^T \log f(y_t \mid y_{t-1}, x_t; \theta). \quad (3)$$

For a candidate structural parameter vector β , simulate M artificial samples and estimate the same auxiliary model on the simulated data:

$$\tilde{\theta}_{T,M}(\beta) = \arg \max_{\theta \in \Theta} \frac{1}{MT} \sum_{m=1}^M \sum_{t=1}^T \log f(\tilde{y}_t^m(\beta) \mid \tilde{y}_{t-1}^m(\beta), x_t; \theta). \quad (4)$$

In many applications, the auxiliary estimator is not a maximum-likelihood estimator. It may instead be OLS, IV, a VAR, a probit, a linear probability model, a duration model, a transition matrix, or a vector of regression and decomposition coefficients. The notation above is useful because it nests the classical likelihood-based presentation. More generally, write

$$\hat{\theta}_T = A_T(y, x), \quad \tilde{\theta}_{T,M}(\beta) = A_{T,M}(\tilde{y}(\beta), x),$$

where A denotes the auxiliary estimation procedure.

4 The binding function

The central population object is the binding function:

$$h(\beta) = \text{plim}_{T \rightarrow \infty} \tilde{\theta}_{T,M}(\beta), \quad (5)$$

with M fixed or large enough that simulation error is asymptotically controlled. The binding function maps structural parameters into the population auxiliary parameters implied by the structural model. The observed auxiliary estimator converges to

$$\hat{\theta}_T \xrightarrow{p} \theta_0. \quad (6)$$

If the structural model is correctly specified at β_0 , then the observed data are generated by the same mechanism used in simulation at β_0 , and therefore

$$\theta_0 = h(\beta_0). \quad (7)$$

Identification requires that (7) have a unique solution. In the exactly identified case, $p = k$, this is a local invertibility condition on h . In the overidentified case, $p > k$, the condition is that $h(\beta)$ be closest to θ_0 at a unique β_0 under the chosen population metric.

A rank condition gives local identification. If h is differentiable at β_0 , define the $p \times k$ Jacobian

$$H_0 = \frac{\partial h(\beta_0)}{\partial \beta'}. \quad (8)$$

Local identification requires

$$\text{rank}(H_0) = k. \quad (9)$$

This condition says that the auxiliary statistics must move in linearly independent directions when each structural parameter is perturbed. It is the indirect-inference analogue of a relevance condition.

5 Wald estimator

The most common implementation is the Wald version. Let W_T be a positive definite weighting matrix converging in probability to W . The estimator solves

$$\hat{\beta}_W = \arg \min_{\beta \in \mathcal{B}} Q_T(\beta), \quad Q_T(\beta) = [\hat{\theta}_T - \tilde{\theta}_{T,M}(\beta)]' W_T [\hat{\theta}_T - \tilde{\theta}_{T,M}(\beta)]. \quad (10)$$

If $p = k$ and the mapping is numerically well behaved, the estimator approximately solves

$$\hat{\theta}_T = \tilde{\theta}_{T,M}(\hat{\beta}_W). \quad (11)$$

If $p > k$, the criterion generally remains positive at the optimum. The minimized value is then informative about overidentifying restrictions.

The Wald form is attractive because it is transparent: the researcher can report the observed auxiliary estimate, the simulated auxiliary estimate at $\hat{\beta}_W$, and the contribution of each component to the criterion.

6 Likelihood-ratio and score versions

When the auxiliary model is likelihood-based, two other classical metrics are available.

6.1 Likelihood-ratio version

The likelihood-ratio criterion compares the observed-data auxiliary likelihood evaluated at the unrestricted auxiliary estimate $\hat{\theta}_T$ to the same likelihood evaluated at the simulated-model-implied auxiliary estimate $\tilde{\theta}_{T,M}(\beta)$:

$$\hat{\beta}_{LR} = \arg \min_{\beta \in \mathcal{B}} \left\{ \frac{1}{T} \sum_{t=1}^T \log f(y_t | y_{t-1}, x_t; \hat{\theta}_T) - \frac{1}{T} \sum_{t=1}^T \log f(y_t | y_{t-1}, x_t; \tilde{\theta}_{T,M}(\beta)) \right\}. \quad (12)$$

The first term does not depend on β . The criterion is nonnegative by construction when $\hat{\theta}_T$ maximizes the auxiliary likelihood. Intuitively, the structural parameters impose restrictions on the auxiliary parameter through the binding function; the LR estimator chooses the restrictions that imply the smallest loss in auxiliary likelihood.

6.2 Score or LM version

Let the auxiliary score be

$$s_t(y; \theta) = \frac{\partial}{\partial \theta} \log f(y_t | y_{t-1}, x_t; \theta). \quad (13)$$

The score version evaluates the simulated auxiliary score at the observed-data auxiliary estimate:

$$S_{T,M}(\beta) = \frac{1}{MT} \sum_{m=1}^M \sum_{t=1}^T s_t(\tilde{y}^m(\beta); \hat{\theta}_T), \quad (14)$$

then solves

$$\hat{\beta}_{LM} = \arg \min_{\beta \in \mathcal{B}} S_{T,M}(\beta)' V_T S_{T,M}(\beta), \quad (15)$$

where V_T is positive definite. The observed-data score is zero at $\hat{\theta}_T$ by the first-order condition of the auxiliary MLE. Hence the score version chooses β so that the simulated score, evaluated at the observed auxiliary estimate, is close to zero.

The practical advantage is that the auxiliary model need not be re-estimated for every candidate β . This matters when the auxiliary model is nonlinear, high-dimensional, or computationally expensive.

7 Consistency

The consistency argument is simple but useful. Suppose:

1. $\mathcal{B}$ is compact and β_0 is in its interior.
2. $\hat{\theta}_T \xrightarrow{p} \theta_0$.
3. $\tilde{\theta}_{T,M}(\beta)$ converges uniformly in probability to $h(\beta)$:

$$\sup_{\beta \in \mathcal{B}} \|\tilde{\theta}_{T,M}(\beta) - h(\beta)\| \xrightarrow{p} 0.$$

4. $W_T \xrightarrow{p} W$, with W positive definite.
5. The population criterion

$$Q(\beta) = [\theta_0 - h(\beta)]' W [\theta_0 - h(\beta)]$$

has a unique minimizer at β_0 .

Then

$$\hat{\beta}_W \xrightarrow{p} \beta_0. \quad (16)$$

The identifying content is contained in the final condition. Simulation and optimization cannot compensate for an auxiliary model that does not distinguish the structural parameters. If $h(\beta_1) = h(\beta_2)$ for two distinct parameter values, then those values are observationally equivalent from the vantage point of the chosen auxiliary model.

8 Asymptotic distribution

Let

$$H_0 = \frac{\partial h(\beta_0)}{\partial \beta'}.$$

Assume that

$$\sqrt{T}(\hat{\theta}_T - \theta_0) \Rightarrow N(0, \Omega), \quad (17)$$

where Ω is the asymptotic covariance matrix of the auxiliary estimator on observed data. With M independent simulation paths of the same length and common regularity conditions, simulation adds an additional variance component. In the standard independent-simulation case, the leading inflation factor is approximately

$$1 + \frac{1}{M}. \quad (18)$$

Consequently, the Wald estimator has the asymptotic form

$$\sqrt{T}(\hat{\beta}_W - \beta_0) \Rightarrow N\left(0, \left(1 + \frac{1}{M}\right) (H_0' W H_0)^{-1} H_0' W \Omega W H_0 (H_0' W H_0)^{-1}\right). \quad (19)$$

With the efficient weighting matrix $W = \Omega^{-1}$, this reduces to

$$\sqrt{T}(\hat{\beta}_W - \beta_0) \Rightarrow N\left(0, \left(1 + \frac{1}{M}\right) (H_0' \Omega^{-1} H_0)^{-1}\right). \quad (20)$$

Equation (18) gives a practical rule: increasing M reduces simulation-induced uncertainty. If $M = 1$, the factor is 2 in variance, or $\sqrt{2}$ in standard errors. If $M = 10$, the standard-error inflation is $\sqrt{1.1} \approx 1.049$, usually small relative to other sources of uncertainty.

In empirical panels with clustering, serial dependence, or cross-sectional dependence, Ω should be estimated using the same dependence structure that would be used for the auxiliary estimator itself. The indirect-inference covariance formula then treats the auxiliary covariance as an input, rather than pretending that observations are independent.

9 Choice of weighting matrix

The weighting matrix determines which auxiliary discrepancies matter most in finite samples. Three common choices are:

1. **Identity weighting.** Set $W = I_p$. This is simple and useful for debugging, but it is scale-dependent. A statistic measured in large units may dominate the objective.
2. **Diagonal variance weighting.** Set $W = \text{diag}(\hat{\Omega})^{-1}$. This standardizes each auxiliary statistic by its estimated variance while avoiding instability from estimating a full covariance matrix.
3. **Efficient weighting.** Set $W = \hat{\Omega}^{-1}$. This is asymptotically efficient under the usual conditions, but it can be numerically fragile when p is large, auxiliary statistics are highly correlated, or $\hat{\Omega}$ is ill-conditioned.

A robust empirical workflow is to estimate the model first with identity or diagonal weighting, inspect which moments are driving the fit, and only then consider a full efficient weighting matrix. If the efficient-weighting estimate differs sharply from the diagonal-weighting estimate, the difference often reflects covariance estimation noise or a near-singular auxiliary covariance matrix rather than a deep economic result.

10 Relationship to simulated method of moments

The simulated method of moments (SMM) is a special case of indirect inference (Duffie and Singleton, 1993). Let $\hat{m}_T$ be a vector of empirical moments and let $\tilde{m}_{T,M}(\beta)$ be the same vector computed from simulated data. The SMM estimator solves

$$\hat{\beta}_{SMM} = \arg \min_{\beta \in \mathcal{B}} [\hat{m}_T - \tilde{m}_{T,M}(\beta)]' W_T [\hat{m}_T - \tilde{m}_{T,M}(\beta)]. \quad (21)$$

This is identical to the Wald indirect-inference estimator if the auxiliary estimator is simply the moment vector $A_T(y, x) = \hat{m}_T$.

The distinction is therefore conceptual rather than mechanical:

- SMM usually starts from hand-chosen moments: means, variances, covariances, transition rates, elasticities, or shares.
- Indirect inference starts from an auxiliary econometric model: OLS, IV, VAR, probit, duration model, auxiliary likelihood, or an entire reduced-form system.

The indirect-inference language is especially useful when the targeted objects are themselves estimates from an econometric procedure. Examples include regression coefficients, policy functions, impulse responses, forecast-error regressions, and conditional choice probabilities.

11 Auxiliary model design

The auxiliary model is not a decorative component. It is the source of identification and the main determinant of the estimator's credibility.

11.1 Desiderata

A good auxiliary model should satisfy five criteria.

1. **Feasibility.** It must be estimable on both observed and simulated data under the same sample restrictions.
2. **Identification.** Its parameters must respond differently to different structural parameters.
3. **Economic relevance.** It should summarize mechanisms that the structural model is designed to explain.
4. **Statistical stability.** Auxiliary estimates should not be too noisy, too collinear, or dominated by rare events.
5. **Replicability.** The same code should compute the auxiliary statistics in the observed and simulated samples.

11.2 Examples

Time-series structural models. A nonlinear DSGE model may be estimated by choosing parameters so that a VAR estimated on simulated output and consumption matches a VAR estimated on observed aggregate data (Smith, 1993). The auxiliary VAR does not have to be

the true law of motion. Its role is to summarize persistence, comovement, and volatility in a tractable way.

Discrete-choice models. A structural discrete-choice model may imply choice probabilities that are costly to evaluate exactly. An auxiliary linear probability model or multinomial model can summarize how observed choices vary with states. Indirect inference then chooses structural utility parameters so that simulated choices reproduce the same auxiliary choice patterns (Keane and Smith, 2003).

Dynamic oligopoly. In dynamic industry models, state variables, strategic interaction, entry, exit, and investment often make full likelihood estimation impractical. Auxiliary objects may include conditional choice probabilities, transition rates across market structures, investment regressions, or the response of entry to demand states. Applications to dynamic industry structure emphasize that the computational burden often lies in solving or approximating the dynamic game repeatedly (Collard-Wexler, 2013).

Consumption and income dynamics. In life-cycle consumption models, auxiliary regressions linking consumption changes, consumption levels, and income histories can be designed to distinguish persistent income shocks from predictable heterogeneity in income profiles. Such designs use economic choices to infer latent income risk and insurance mechanisms (Guvenen and Smith, 2014).

12 Overidentification and specification testing

When $p > k$, the model generally cannot match all auxiliary statistics exactly. Let

$$\hat{Q}_T = Q_T(\hat{\beta}_W). \quad (22)$$

A large minimized criterion indicates that the structural model cannot reproduce the auxiliary patterns jointly. With efficient weighting and standard regularity conditions, a scaled version of the minimized criterion has a chi-square limiting distribution with $p - k$ degrees of freedom:

$$T\hat{Q}_T \Rightarrow \chi_{p-k}^2, \quad (23)$$

up to simulation adjustments and normalization conventions. In practice, the diagnostic value of the test is often more important than its literal p -value. A transparent report should show:

- each observed auxiliary statistic $\hat{\theta}_j$;
- the simulated counterpart $\tilde{\theta}_j(\hat{\beta})$;
- the standardized discrepancy;
- the contribution of each discrepancy to the objective.

This decomposition prevents the estimator from becoming a black box. It also reveals whether model failure is broad or concentrated in a small subset of moments.

13 Incidental parameters and bias correction

Indirect inference can also be used as a bias-correction device. Consider a fixed-effects estimator in a short panel with individual effects. The direct estimator of a structural parameter may be biased because the number of nuisance parameters grows with the cross-sectional dimension, while the time dimension remains fixed. This is the classical incidental-parameters problem (Neyman and Scott, 1948).

The indirect-inference correction proceeds as follows. First, estimate the biased auxiliary estimator on the observed panel. Second, for a candidate structural parameter, simulate panels with the same dimensions and estimate the same biased auxiliary estimator on each simulated panel. Third, choose the structural parameter so that the simulated biased estimator matches the observed biased estimator. The logic is not to remove the bias analytically. It is to reproduce the finite-sample bias inside the simulation and invert it.

This use of indirect inference is especially attractive when the analytical bias formula is difficult to derive but the finite-sample estimation procedure can be replicated exactly in simulation.

14 Implementation algorithm

A practical implementation can be organized as follows.

1. **Define the observed sample.** Fix the sample restrictions, units, time period, transformations, trimming rules, and missing-data conventions. These choices must also be applied to simulated data.
2. **Estimate the auxiliary model on observed data.** Store $\hat{\theta}_T$, its covariance estimator $\hat{\Omega}_T$, and any auxiliary objects needed for replication.
3. **Specify structural primitives.** Define the state variables, controls, shocks, transition equations, equilibrium conditions, initial conditions, and parameter bounds.
4. **Draw simulation shocks.** Generate M shock histories. Hold them fixed across parameter evaluations unless there is a specific reason to resample.
5. **Simulate the model.** Given β , solve the model if needed, generate artificial data, and impose the same observation structure as in the real sample.
6. **Estimate the auxiliary model on simulated data.** Compute $\tilde{\theta}_{T,M}(\beta)$ using exactly the same auxiliary procedure as in the observed data.
7. **Evaluate the criterion.** Compute

$$Q_T(\beta) = [\hat{\theta}_T - \tilde{\theta}_{T,M}(\beta)]' W_T [\hat{\theta}_T - \tilde{\theta}_{T,M}(\beta)].$$

8. **Optimize.** Use a global or derivative-free method to find a stable region, followed by a local optimizer for refinement. If the objective is noisy or nonsmooth, derivative-free methods may be preferable throughout.
9. **Conduct inference.** Estimate the Jacobian H_0 numerically or by automatic differentiation if available. Combine it with $\hat{\Omega}_T$ and the simulation factor. As an alternative, use a bootstrap that re-estimates the auxiliary model and re-solves the indirect-inference problem.
10. **Report diagnostics.** Report moment fit, sensitivity to weights, sensitivity to simulation seeds, convergence checks, and local identification diagnostics.

15 Numerical issues

15.1 Nonsmoothness

Simulated objectives can be nonsmooth, especially in discrete-choice, entry-exit, or occasionally binding-constraint models. Nonsmoothness arises because small changes in β may change simulated choices discontinuously. Common remedies include:

- common random numbers;

- smoothing of simulated choice indicators;
- larger simulated samples;
- derivative-free optimization;
- multi-start optimization;
- separating debugging runs from final high-precision runs.

15.2 Weak auxiliary information

If the auxiliary statistics are weakly sensitive to a parameter, the objective will be flat in that direction. Numerically, this appears as unstable estimates, large standard errors, or frequent boundary solutions. A finite-difference Jacobian is useful:

$$\hat{H}_j = \frac{\tilde{\theta}(\hat{\beta} + \epsilon e_j) - \tilde{\theta}(\hat{\beta} - \epsilon e_j)}{2\epsilon}. \quad (24)$$

Small or collinear columns of $\hat{H}$ indicate weak local identification.

15.3 Simulation size

A large simulated sample reduces simulation noise but increases runtime. There are two distinct choices:

- increase M , the number of independent simulation paths;
- increase the size of each simulated panel or time series.

The first choice directly reduces the standard simulation factor $1 + 1/M$. The second can also help when the simulated auxiliary estimator is noisy because simulated rare events are poorly represented.

15.4 Boundary estimates

Boundary estimates may have three interpretations:

1. the true parameter lies near the boundary;
2. the auxiliary model does not identify the parameter;
3. the model is misspecified and the optimizer uses the boundary to compensate.

A boundary solution should trigger sensitivity checks, not immediate substantive interpretation.

16 Reporting standards

A technically complete indirect-inference application should report at least:

1. the structural parameter vector and parameter bounds;
2. the auxiliary model and the exact auxiliary statistics used;
3. the observed auxiliary estimates and their covariance matrix or standard errors;
4. the number of simulations and the simulation design;
5. the weighting matrix;
6. the optimizer and convergence criteria;
7. the estimated structural parameters and standard errors;

8. the observed-versus-simulated auxiliary fit;
9. overidentification or specification diagnostics when $p > k$;
10. robustness to weighting, simulation seeds, and auxiliary specification.

A useful table has the following structure:

Auxiliary statistic	Data	Simulation	Difference	Std. difference
Statistic 1	$\hat{\theta}_1$	$\tilde{\theta}_1(\hat{\beta})$	$\hat{\theta}_1 - \tilde{\theta}_1$	z_1
Statistic 2	$\hat{\theta}_2$	$\tilde{\theta}_2(\hat{\beta})$	$\hat{\theta}_2 - \tilde{\theta}_2$	z_2
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

This table is often more informative than the structural parameter table alone.

17 Common pitfalls

1. **Matching too many weak moments.** A large vector of noisy auxiliary statistics can degrade identification and make the weighting matrix unstable.
2. **Using different procedures in data and simulation.** The auxiliary estimator must be replicated exactly. Different filters, transformations, or fixed effects break the logic of the method.
3. **Ignoring simulation error.** A small number of simulations may produce unstable estimates and misleading standard errors.
4. **Treating auxiliary fit as structural truth.** A good fit to selected auxiliary statistics does not imply that the model is correct along untested dimensions.
5. **Overinterpreting overidentification tests.** In rich structural models, formal tests often reject because no model is literally true. The location and economic meaning of the misfit usually matter more than the binary reject/fail-to-reject outcome.
6. **Choosing auxiliary statistics only because they are easy.** Ease of computation is not sufficient. The auxiliary model must be informative about the structural mechanisms.

18 Conclusion

Indirect inference is best understood as an inversion of a simulated binding function. The auxiliary model compresses the data into a set of informative empirical features. The structural model is simulated repeatedly, and its parameters are chosen so that the simulated auxiliary features match the observed ones. The method is flexible because the auxiliary model need not be correctly specified and because the structural likelihood need not be tractable. That flexibility is also its main danger: identification and credibility depend on whether the auxiliary model captures the mechanisms that the structural model is supposed to discipline.

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